

Theory Week 5

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Orga

Mechanics 3 CAB G59
WhatsApp group

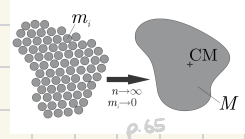


5	14/10/24	kinematics of rigid bodies in 2D, examples	3-3.1.1
	16/10/24	kinematics of rigid bodies in 3D, linear momentum balance	3.1.1-3.2.1
exercise 5: particle impact, particles with variable mass			

3. Rigid Bodies

Def.: $\square$ a **rigid body** is undeformable

$\square$ collection of infinitely small particles packed together infinitely close, with a constant distance between the particles.



3.1 Kinematics

total mass $\leftarrow M = \lim_{\Delta m_i \rightarrow 0} \sum_{i=1}^n \Delta m_i = \int_B dm = \int_B \rho dV$

where

mass density $\leftarrow \rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV}$

pos. of CoM

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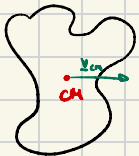
$$\mathbf{r}_{CM}(t) = \frac{1}{M} \int_B \mathbf{r}(t) dm = \frac{1}{M} \int_B \mathbf{r}(t) \rho dV$$

Motion of rigid bodies

Translation

&

Rotation

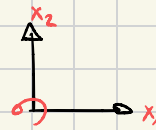


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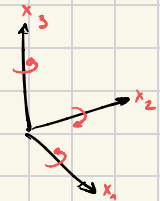


DOFs:

2D:



3D:



2D is described by **3 degrees of freedom**, including translations in the two in-plane directions, e.g., x_1 and x_2 , and an in-plane rotation φ about the out-of-plane x_3 -axis.

3D is described by **6 degrees of freedom**, including translations in the three x_i -directions as well as three rotations about the three x_i -axes.

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$\rightarrow$ Each point P on body can be described uniquely by $\mathbf{r}_P(t)$

therefore if A & B $\in$ body : $\square \mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{AB}$

$\square$ velocity and acceleration transfer formulae & theorem of projected velocities:

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{AB}$$

$$(\mathbf{v}_A - \mathbf{v}_B) \cdot \mathbf{r}_{AB} = 0$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\omega}} \times \mathbf{r}_{AB} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{AB})$$

Same as Mech 1 with Mazza!

Def.: $\square$ **Instantaneous center of rotation (ICR)** $\rightarrow \mathbf{v}_P = \mathbf{v}_{ICR} = 0$

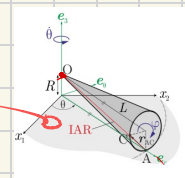
$\hookrightarrow$ ICR can be inside or outside of body!

$\square$ **Instantaneous axis of rotation (IAR)** $\rightarrow \forall P \in \text{IAR} = 0$

$\Rightarrow$

If two distinct points on the same (rotating) body have zero velocity, then the body's IAR goes through those two points.

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3.2 Kinetics

LMB

balance of linear momentum if $M = \text{const.}$:

$$\sum_i \mathbf{F}_i^{\text{ext}} = \dot{\mathbf{P}} = \frac{d}{dt}(M\mathbf{v}_{\text{CM}}) \quad \Rightarrow \quad \sum_i \mathbf{F}_i^{\text{ext}} = M\mathbf{a}_{\text{CM}}$$

□ as always, if there are no $\mathbf{F}^{\text{ext}}$ that are acting on our system (here the rigid body)

$\Rightarrow$ conservation of lin. mom.

$$\Rightarrow \frac{d}{dt} \underline{P} = 0 \rightarrow M \cdot \underline{v}_{\text{CM}} = \text{const}$$